

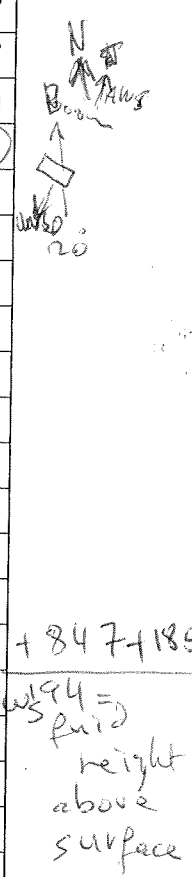
AWS checklist and maintenance guide

Station name	QAS L
People present	JOE, MAU, Pilot Simon instructed by: MAU
Purpose	standard maintenance / installation / take-down / quick visit / emergency
Date	23/8/14
Transport	helicopter / fixed wing / snow mobile / dog sleds / on foot from:
Weather	warm / cold / very cold / no wind / some wind / rain / cloud cover %:

Notes:

Metadata before maintenance (if establishing a new AWS: skip this page)

Time difference between logger clock and UTC	1m55s	+/-
Adjust logger time to UTC (be sure PC runs on UTC time)		Y/N
Download data to PC and/or change CF Card (wait for green light)		Y/N
Name of logger program	Prm.2206bin07.csl	
Photo or screen dump of values in fast scan mode		Y/N
Latitude (dd mm.mmm)		
Longitude (dd mm.mmm)		
Photos of tripod, sensors, logger box wiring etc. (more is better)		Y/N
Mast tilt in boom direction (+ if radiometer tilting down)		°
Mast tilt across boom looking from radiometer (- if clockwise)		°
Boom direction relative to north	20	° true/magn
Box on wind sensor exactly towards mast? If not, measure direction.		Y/N
Radiometer aligned with mast and inclinometer? If not, measure.		Y/N
Temperature / humidity height (bottom of casing – surface)		cm
Sonic ranger on AWS height (membrane – surface)	275	cm
Wind sensor height (center of propeller – surface)		cm
Sonic ranger on stakes height (membrane – surface)		cm
Free length of stake – outer, holding boom with sonic ranger		cm
Free length of stake – middle		cm
Free length of stake – outer, holding boom without sonic ranger		cm
Free length of ablation hose on/above the ice surface		cm
Alt: depth of ablation hose using markings on hose	344	cm
Height of ablation hose fluid level above the ice surface	344 + 460 + 1000	cm
Vertical difference in surface height (at station – at hose) (approx)	hose hole 29 cm higher than station	cm
Length of thermistor string on surface from surface marking	600	cm
Height of surface irregularities	20	cm
Stake assembly still standing?	Y/N / more or less	
Apparent damage to sensors? If yes, which?		Y/N



Notes:

leaning stake assembly, underestimating height

see photos! JEB, MAV

5R50

$\alpha = 152^\circ$

200-34 = 166 or 366!

278 cm

378

approx!

847 + 185 = 1032 = fluid height above surface

Part	Old part number	New part number, if replaced	Potential maintenance tasks
Radiometer		<u> </u>	- Aligned with mast: Y/N - Clean: Y/N
Inclinometer		<u> </u>	- Aligned with radiometer within 0.5°: Y/N
Satellite antenna		<u> </u>	
Wind sensor		<u> </u>	- Box on sensor exactly towards mast (south): Y/N
Temperature / humidity assembly			
T/hum assembly casing			- Fan spins and sounds OK: Y/N
T/hum assembly bracket			
Temperature probe in T/hum assembly			
HygroClip in T/hum assembly			- 40 °C offset in HygroClip temperature: Y/N
Sonic ranger on AWS		<u> </u>	- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N
Solar panel		<u> </u>	- Output OK: Y/N - Clean: Y/N
GPS antenna		<u> </u>	- Stuck to top of logger enclosure: Y/N
Data logger		<u> </u>	- New internal battery in old logger: Y/N - New operating system: Y/N
Card reader		<u> </u>	- LED green of flashing orange (active): Y/N
Multiplexer		<u> </u>	- Clicking as it should: Y/N
Iridium modem IMEI number		<u> </u>	- Transmissions tested: Y/N - Problems: Y/N
Barometer		<u> </u>	
Sonic ranger on stakes		<u> </u>	- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N - In danger of being buried next winter: Y/N
Thermistor string		<u> </u>	- Old string left in ice <u>Y/N/cut</u> <i>nearly out</i>
Ablation hose assembly			- Old hose left in ice: Y/N/cut - Air in hose: Y/N/removed - Bladder 50% full: Y/N / after emptying/filling
Logger enclosure	- Replaced (including everything inside): Y/N - New vent filter in logger enclosure: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Battery box	- New box with new batteries: Y/N - Old box with new batteries: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Stakes	- New ones drilled in: <u>Y/N</u> - Length: 6 / 8 / 10 m - Caps in bottom end to reduce melt-in: Y/N		
Tripod	- New parts: Y/N - Boom direction: true north-south: Y/N		

Metadata after maintenance / establishment

Time logger clock set (to UTC)?	Y/N
Name of logger program	2014d
Photo or screen dump of values in fast scan mode	☒ Y ☐ N
Latitude (dd mm.mmm)	
Longitude (dd mm.mmm)	
Photos of tripod, sensors, logger box wiring etc. (more is better)	☒ Y ☐ N
Mast tilt in boom direction (+ if radiometer tilting down)	+ 1 °
Mast tilt across boom looking from radiometer (- if clockwise)	level °
Boom direction relative to north (radiometer should point to true south)	5 ° true (magn)
Box on wind sensor exactly towards mast? If not, change.	3° W of S ☒ Y ☐ N
Radiometer aligned with mast and inclinometer? If not, align.	☒ Y ☐ N
Temperature / humidity height (bottom of casing – surface)	268.335 cm
Sonic ranger on AWS height (membrane – surface)	274.330 cm
Wind sensor height (center of propeller – surface)	320 cm
Sonic ranger on stakes height (membrane – surface)	224 cm
Free length of stake – outer, holding boom with sonic ranger	310 cm
Free length of stake – middle	320 cm
Free length of stake – outer, holding boom without sonic ranger	320 cm
Free length of ablation hose on/above the ice surface	cm
Alt: depth of ablation hose using markings on hose	cm
Height of ablation hose fluid level above the ice surface	194 cm
Vertical difference in surface height (at station – at hose) (approx)	cm
Length of thermistor string on surface from surface marking	26.5 cm
Height of surface irregularities	cm

Notes:

Thermistor string top marker red tape with yellow strip marking 0 m, made by Babis?